

MATH201 · Differential Equations

Week 2 lecture notes

Families of curves (continued)

Example 1.1.10 — $y = C_1x + C_2e^{-x}$

$y' = C_1 - C_2e^{-x}$ and $y'' = C_2e^{-x}$, so $C_1 = y' + y''$ and

$$y = x(y' + y'') + y''.$$

Example 1.1.11 — $y = x^2 + C_1e^x + C_2e^{-x}$

Since $y'' = 2 + C_1e^x + C_2e^{-x}$, subtraction eliminates both constants:

$$y = x^2 + y'' - 2, \quad \text{equivalently} \quad y'' - y = 2 - x^2.$$

First-order differential equations

Separable differential equations

A first-order equation can be written as $d_y y = f(x, y)$, or in differential form as $M(x, y)dx + N(x, y)dy = 0$ (with $f = -\frac{M}{N}$ where $N \neq 0$). It is **separable** if it can be rearranged into $P(x)dx + Q(y)dy = 0$: collect the y terms with dy and the x terms with dx . Direct integration gives

$$\int P(x)dx + \int Q(y)dy = C.$$

When dividing by a factor involving x or y , check the excluded values against the equation.

Example 1.2.1

Solve $x(y^2 - 1)dx - y(x^2 - 1)dy = 0$ with $y(2) = 2$. On the region $x > 1, y > 1$ containing the initial point, separate:

$$\frac{xdx}{x^2 - 1} - \frac{ydy}{y^2 - 1} = 0, \quad \int \frac{xdx}{x^2 - 1} = \int \frac{ydy}{y^2 - 1}.$$

For the left integral, set $u = x^2 - 1$, so $du = 2xdx$:

$$\int \frac{xdx}{x^2 - 1} = \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln(x^2 - 1).$$

Similarly, set $v = y^2 - 1$, $dv = 2ydy$ on the right. Thus

$$\ln(x^2 - 1) - \ln(y^2 - 1) = C, \quad \frac{x^2 - 1}{y^2 - 1} = e^C = K.$$

Since e^C is also a constant, it may be renamed C ; here we use K to distinguish it, with $K > 0$ on this region. The initial condition gives $K = 1$, so $x^2 = y^2$. Since $x, y > 1$,

$$y = x.$$

The differential equation also has the constant solutions $y(x) = 1$ and $y(x) = -1$, excluded by division by $y^2 - 1$. Neither satisfies $y(2) = 2$.

Example 1.2.2

Solve $(1 + x^2 + y^2 + x^2y^2)dy = y^2dx$ with $y(0) = -1$. Group and factor:

$$1 + x^2 + y^2 + x^2y^2 = x^2(1 + y^2) + (1 + y^2) = (1 + x^2)(1 + y^2).$$

For $y \neq 0$, separate:

$$\frac{(1 + y^2)dy}{y^2} = \frac{dx}{1 + x^2}.$$

Integration yields $y - \frac{1}{y} = \arctan x + C$. From $y(0) = -1$, $C = 0$; therefore

$$y - \frac{1}{y} = \arctan x, \quad y(0) = -1.$$

The constant solution $y(x) = 0$ satisfies the equation but is lost when dividing by y^2 . It does not satisfy $y(0) = -1$.

Example 1.2.3

Solve $(x^2 - 2x + 1)dy - (y^2 + 2y + 1)dx = 0$ with $y(2) = 1$. The factors are $(x - 1)^2$ and $(y + 1)^2$. For $x \neq 1$ and $y \neq -1$, division and integration give

$$\int \frac{dy}{(y + 1)^2} - \int \frac{dx}{(x - 1)^2} = 0.$$

Thus $-\frac{1}{y+1} + \frac{1}{x-1} = C$. The initial condition gives $C = \frac{1}{2}$, so

$$-\frac{1}{y+1} + \frac{1}{x-1} = \frac{1}{2}.$$

The constant solution $y(x) = -1$ also satisfies the differential equation and is excluded by this division. It does not satisfy $y(2) = 1$.

Example 1.2.5

Solve $\tan^2 y dy = \sin^3 x dx$ with $y(0) = 0$. The equation requires $\cos y \neq 0$. Using $\tan^2 y = \sec^2 y - 1$,

$$\int \tan^2 y dy = \tan y - y.$$

For the right integral, write $\sin^3 x = \sin x(1 - \cos^2 x)$ and set $u = \cos x$, $du = -\sin x dx$:

$$\int \sin^3 x dx = -\int (1 - u^2) du = -u + \frac{u^3}{3} = -\cos x + \frac{\cos^3 x}{3}.$$

Hence $\tan y - y = -\cos x + \frac{\cos^3 x}{3} + C$. At $(0, 0)$, $C = \frac{2}{3}$, so

$$\tan y - y = -\cos x + \frac{\cos^3 x}{3} + \frac{2}{3}.$$

Example 1.2.6

Solve $xy^2(1+x)dy + (1+y^3)dx = 0$ with $y(1) = 1$. Division by $x(1+x)(1+y^3)$ requires $x \neq 0$, $x \neq -1$, and $y \neq -1$:

$$\frac{y^2 dy}{1+y^3} + \frac{dx}{x(1+x)} = 0.$$

Set $u = 1 + y^3$, so $du = 3y^2 dy$; for the other integral use partial fractions:

$$\int \frac{y^2 dy}{1+y^3} = \frac{1}{3} \int \frac{du}{u} = \frac{1}{3} \ln|1+y^3|, \quad \frac{1}{x(1+x)} = \frac{1}{x} - \frac{1}{1+x}.$$

$$\int \frac{dx}{x(1+x)} = \ln|x| - \ln|1+x| = \ln\left|\frac{x}{x+1}\right|.$$

Hence $\frac{1}{3} \ln|1+y^3| + \ln\left|\frac{x}{x+1}\right| = \ln K$, or $\frac{x(1+y^3)^{\frac{1}{3}}}{x+1} = K$ on the branch of the initial point. The condition gives $K = 2^{-\frac{2}{3}}$. Cubing yields

$$\frac{x^3(1+y^3)}{(x+1)^3} = \frac{1}{4}.$$

The differential equation also admits $y(x) = -1$, lost when dividing by $1+y^3$. It does not satisfy $y(1) = 1$.

Example 1.2.9

Solve $y' = x \sin \frac{x}{\tan y} y$ with $y(0) = \pi$. The equation requires both $\sin y \neq 0$ and $\cos y \neq 0$, so that $\tan y$ is defined and nonzero. Separation gives $\tan y dy = x \sin x dx$. For the left integral, set $t = \cos y$, $dt = -\sin y dy$:

$$\int \tan y dy = \int \frac{\sin y}{\cos y} dy = - \int \frac{dt}{t} = -\ln|\cos y|.$$

For the right integral, integration by parts with $u = x$, $dv = \sin x dx$, $du = dx$, and $v = -\cos x$ gives

$$\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x.$$

Therefore

$$-\ln|\cos y| = -x \cos x + \sin x + C.$$

At $y = \pi$, $\tan y = 0$, so the differential equation is undefined at the initial point. For a solution with $\lim_{x \rightarrow 0^+} y(x) = \pi$, taking the limit in the integrated relation gives $C = 0$. Define $A(x) = \sin x - x \cos x$; then

$$|\cos y| = e^{-A(x)}.$$

Near $y = \pi$, $\cos y < 0$, so $\cos y = -e^{-A(x)}$. Since $A(0) = 0$ and $A'(x) = x \sin x > 0$ for $0 < x < \pi$, we have $A(x) > 0$ there. There are two branches approaching π from opposite sides:

$$y_-(x) = \pi - \arccos(e^{-A(x)}), \quad y_+(x) = \pi + \arccos(e^{-A(x)}), \quad 0 < x < \pi.$$

Both satisfy the equation on this open interval and tend to π as $x \rightarrow 0^+$. For $-\pi < x < 0$, $A(x) < 0$, so $e^{-A(x)} > 1$: no real solution of the $C = 0$ relation exists there. Hence there is no real solution on $-\pi < x < 0$ that tends to π as $x \rightarrow 0^-$.

Both branches extend continuously to $y(0) = \pi$. The initial value problem has no classical solution at $x = 0$, because the right-hand side of the differential equation is undefined there.